

Ceramic Glaze Color-Response Test Program

Colorant Layout (one 5×5 tile) — % of dry base weight.

Colorant	1	2	3	4	5
Copper carbonate	0.5	1	2	4	6
Red iron oxide (Fe ₂ O ₃)	1	2	4	8	12
Manganese dioxide (MnO ₂)	1	2	4	8	12
Chrome oxide (Cr ₂ O ₃)	0.2	0.4	0.6	1	2
Cobalt oxide (Co ₃ O ₄)	0.2	0.4	0.6	1	2

Base Treatments (tiles 1–5)

Tile	Base treatment (every spot)
1	Base glaze (unmodified)
2	Base + 5% zircopax
3	Base + 10% zircopax
4	Base + 4% rutile
5	Base + 5% zircopax + 4% rutile

Triaxial Tests (tiles 6–10) — the 10 unique 3-colorant blends ($\binom{5}{3} = 10$), two per tile. Corners: A = top-left, B = top-right, C = bottom-left, D = bottom-right. The shared line blend runs A→D, so the two triaxials on each tile are A-B-D and A-C-D (corners = level-5 loading).

Tile	A	B	C	D
6	Copper	Manganese	Chrome	Red iron oxide
7	Copper	Red iron oxide	Manganese	Cobalt
8	Copper	Manganese	Cobalt	Chrome
9	Red iron oxide	Chrome	Cobalt	Manganese
10	Chrome	Red iron oxide	Manganese	Cobalt